

High Performance Tangential Flow Filtration

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Abstract: Conventional tangential flow filtration (TFF) has traditionally been limited to separation of solutes that differ by about ten-fold in size. Wide pore-size distributions, membrane fouling, and concentration polarization phenomena have commonly been cited as reasons for this limitation. The use of TFF in the biotechnology industry has therefore been restricted to cell-protein, virus-protein, and protein-buffer separations. A multi-disciplinary team with industrial and academic members was formed to overcome these limitations and enable protein-protein separations using High Performance TFF (HPTFF) systems. Pore-size distributions have been improved with the development of new membrane formulation and casting techniques. Membrane fouling has been controlled by operating in the transmembrane pressure-dependent regime of the filtrate flux curve and by carefully controlling fluid dynamic start-up conditions. Concentration polarization was exploited to enhance, rather than limit, the resolution of solutes. Concentration polarization has also been controlled by operating a co-current filtrate stream that maintains transmembrane pressure constant along the length of the TFF module. High yields and purification factors were obtained even with small differences in protein sieving. IgG-BSA and BSA monomer-oligomer mixtures have successfully been separated with these systems. HPTFF technology provides a competitive purification tool to complement chromatographic processing of proteins. © 1997 John Wiley & Sons, Inc. *Biotechnol Bioeng* 56: 71–82, 1997.

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INTRODUCTION

Tangential flow filtration (TFF) has been traditionally regarded as a unit operation that is inherently limited to separating solutes differing by more than ten-fold in size (Cher-

kasov and Polotsky, 1996). This has limited the application of TFF in the biotechnology industry to three main applications. Cell-protein separations, such as medium exchange (Arathoon et al., US Patent #5,053,334, 1991), perfusion (Feder and Tolbert, 1983), and protein clarification or cell harvest (Maiorella et al., 1991; van Reis et al., 1991), employ microporous filtration membranes. These membranes effectively separate proteins (1–15 nm) from bacteria (0.5–5 μm), yeast (2–15 μm) and mammalian cells (5–60 μm). Ultrafiltration has been used successfully for protein concentration and buffer exchange (Kurnik et al., 1995) in which molar masses of solutes differ by more than ten-fold. Separations of solutes with 10-fold size ratios have been accomplished in virus filtration applications (DiLeo et al., 1992).

The limitations to successfully applying ultrafiltration in protein separations can be attributed to a number of phenomena. In particular, wide pore-size distributions can significantly limit membrane resolving power (Mochizuki and Zydney, 1993), concentration polarization can reduce selectivity values, and membrane fouling and protein-protein interactions can interfere with separations. Tangential flow velocity can be varied to modify the extent of polarization and in turn, the membrane wall concentrations of solutes. By proper selection of the solute wall concentrations, in combination with the intrinsic sieving coefficients, one can significantly alter the separation characteristics of the fluid/membrane system, as has been demonstrated in both unstirred (Nakatsuka and Michaels, 1992) and stirred cells (Saksena and Zydney, 1994). The same effects can also be seen by varying transmembrane pressure or filtrate flux. Conventional TFF typically operates in the pressure-independent part of the filtrate flux curve. Higher resolution separations can be obtained by operating at transmembrane pressures in the pressure-dependent part of the filtrate flux curve (van Reis, US Patent #5,256,294, 1993; van Reis, US Patent #5,490,937, 1996). Several recent studies have demonstrated, on a limited scale, that protein separations by

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ultrafiltration are possible (Balakrishnan and Agarwal, 1996, Lentsch et al., 1993, Nakao et al., 1988, Millesime et al., 1996, and Sannier et al., 1996).

In some instances, tangential velocity and pressure effects can be combined to yield reverse separations in which the higher molecular weight solute has a larger effective sieving coefficient than the lower molecular weight solute (Jonsson, 1986; Saksena, 1995). This phenomenon can be attributed to a lower diffusion coefficient for the larger solute. This results in a proportionately higher wall concentration for the larger solute. The significantly higher wall concentration results in a larger observed sieving coefficient despite the lower intrinsic sieving coefficient. Protein charge, hydrodynamic volume, and diffusion coefficients can be drastically altered by varying solution pH, ionic strength, and buffering species (Saksena and Zydney, 1994). Protein-protein interactions may also play a major role in protein separations. These interactions can also be modulated through changes in solution environment. High ionic strength may be required in some cases to dissociate proteins that would, otherwise, be inseparable by membrane systems (Ingham et al., 1980).

Membrane fouling has been another major problem in enabling protein separations using membranes (Meireles et al., 1991; Mochizuki and Zydney, 1992). Membrane, protein, buffer chemistry, and fluid dynamics all contribute to membrane fouling (Causerrand et al., 1994). Development of new membrane chemistries, improved mechanical strength for existing low fouling regenerated cellulose membranes, and development of optimal buffer conditions suitable for specific protein/membrane processes all play a role in attaining optimum separations. In addition, it is important to control process start-up procedures with regard to feed rate and filtrate flux.

The formation of a pressure gradient along the length of the feed channel reduces the resolving power of conventional TFF systems. Since the solute sieving coefficients vary over a wide range with pressure (van Eijndhoven et al., 1995), optimum separation conditions are only maintained at a single point along the channel path. This problem can be overcome by operating at a constant and optimum pressure level throughout the filtration system. In systems with large axial pressure drop, this can be accomplished through the use of a co-current filtrate flow to balance the axial pressure drop in the retentate (van Reis, US Patent #5,256,294, 1993). Even in cases where the differences in protein sieving (ΔS) are small, significant yields and purification factors can be obtained by operating processes with a large number of diavolumes. Industrially relevant protein purification can be carried out by HPTFF using high resolution and high filtrate flux membranes in conventional batch diafiltration mode or with open or closed-loop cascade systems.

This article describes the approaches that we have used to overcome the various limitations of membrane systems for protein purification. Results are presented for two different systems: albumin-immunoglobulin and albumin monomer-oligomer separations. Through systematic application of

biochemical, thermodynamic, and chemical engineering principles, we have demonstrated that membrane-based protein purification processes can be developed and effectively implemented at industrial scale. We have also refined a methodology for HPTFF process development that includes membrane selection, buffer optimization, fluid dynamic optimization, and the use of a novel optimization diagram (van Reis and Saksena, 1997).

MATERIALS AND METHODS

Proteins and Buffer Solutions

Human immunoglobulin G (IgG) was purchased from Sigma Chemical Company (Cat #G-4386, St. Louis, MO). Sulfhydryl-modified bovine serum albumin (BSA) was obtained from Miles Corporation (Cat #81-024-2, Diagnostics Division, Kankakee, IL). The BSA preparation was used as obtained from the supplier, and contained approximately 70% monomer and 30% oligomer by high performance liquid chromatography (HPLC) size exclusion chromatography (SEC) analysis. The protein mixtures used in all of the IgG-BSA experiments were filtered through 0.22 μm Millipak™ filters (Millipore Corp., Bedford, MA). Reagent-grade anhydrous sodium acetate, glacial acetic acid, sodium phosphate monobasic, sodium phosphate dibasic, and sodium borate were purchased from Sigma Chemical Company (St. Louis, MO).

Assays

HPLC SEC assays for BSA monomer-oligomer samples were performed on an automated system consisting of a quaternary pump, auto sampler and injector, UV absorbance detector, an interface module, computer and HP Chemstation data acquisition and processing software (Model 1050 HPLC, Hewlett Packard, Palo Alto, CA). The columns used for this assay included a Waters ProteinPak 300 SW (inner diameter = 7.8 mm, length = 300 mm) (Waters Corporation, Milford, MA) and a Superdex 200 HR 10/30 (inner diameter = 10.0 mm, length = 300 mm) (Pharmacia Biotech, Piscataway, NJ). The mobile phase was 13.7 mM sodium phosphate, 145 mM sodium chloride solution at a volumetric flow rate of 1.0 mL min⁻¹. The sample injection volume was 4–100 μL . UV absorbance was measured at a wavelength of either 214 or 280 nm.

An HPLC anion exchange system was used to analyze IgG-BSA samples from the fluid dynamic optimization experiments. An automated system (Waters Corporation, Milford, MA) with a tunable UV absorbance detector (Model 486), auto injector (Model Ultra Wisp 715), controller and pump (Model 600), and Millennium 2.1 data acquisition and processing software was used. The column used for this assay was a Waters Q8HR strong anion exchanger (inner diameter = 5 mm, length = 100 mm). The mobile phase was phosphate buffer obtained from Difco Laboratories

(Cat. #DF2314-15, VWR Scientific, Boston, MA) at a volumetric flow rate of 0.5 mL min^{-1} . The sample injection volume was $50 \text{ }\mu\text{L}$. UV detection was performed at a wavelength of 280 nm .

Capillary electrophoresis assays of IgG-BSA samples from the buffer optimization experiments and the HPTFF separation process were performed on an automated Dionex system (Model CES-1, Dionex Corporation, Sunnyvale, CA). Uncoated silica capillary columns with an inner diameter of $50\text{--}75 \text{ }\mu\text{m}$ and a length of $0.5\text{--}0.75 \text{ m}$ (Cat. #043709, Dionex Corporation, Sunnyvale, CA) were used. The running buffer was 50 or 100 mM sodium borate (Sigma Chemical Corporation, St. Louis, MO) at $\text{pH } 8.5$. The injection volume was controlled by the gravity injection method for a time of $20\text{--}60 \text{ s}$. The potential difference across the capillary was set at $6,000\text{--}20,000 \text{ V}$. UV absorbance was measured at a wavelength of 210 or 280 nm .

Membranes and Membrane Cleaning

All membranes used in this study were obtained from Millipore Corporation (Bedford, MA). The membranes were in the form of stirred cell discs (15 cm^2), Minitan-S[®] sheets (60 cm^2), or Pellicon-2[™] cassettes ($0.1 \text{ m}^2\text{--}0.6 \text{ m}^2$). All nominal molecular weight cut-off values described in this paper are given as the dextran size for which the membrane exhibits an observed retention $R = 0.9$. The retention data were obtained by SEC analysis of feed and filtrate samples of mixed dextran solutions as previously described by Tkacik and Michaels (1991). The three membrane types were: (1) prototype composite regenerated cellulose membranes with nominal molecular weight cut-off values of 90 kD , 130 kD , and 170 kD ; (2) prototype Biomax[™] modified polyethersulfone membranes with nominal molecular weight cut-off values of 150 kD (PBSK) and 300 kD (PBHK); and (3) conventional regenerated cellulose membranes with nominal molecular weight cut-off values of 10 kD (PLGC) and 100 kD (PLHK).

All new membranes were flushed prior to use to remove any residual storage solutions used in the shipping of the membranes and to equilibrate the membrane to the appropriate buffer conditions. If membranes were to be reused they were cleaned with the same procedure as employed for new membranes. Biomax[™] membranes were cleaned using a 525 ppm sodium hypochlorite solution at 50°C for 45 min . Sodium hypochlorite was purchased from either Aldrich (Cat. #2390-5, Milwaukee, WI) or in the form of commercial bleach from Clorox (Oakland, CA). Regenerated cellulose membranes were cleaned using a 0.1 N sodium hydroxide solution and were sanitized using a 300 ppm MinnCare[™] solution, both at ambient temperature. Sodium hydroxide was purchased from J.T. Baker (Phillipsburg, NJ) and MinnCare[™] was purchased from Minntech Corporation (Minneapolis, MN). The Pellicon-2[™] system was cleaned using a feed flux of $324 \text{ L m}^{-2} \text{ h}^{-1}$ (determined as feed flow rate per unit membrane area), and a 40% conversion ratio (filtrate rate/feed rate). The Minitan-S[®] system

was cleaned using a feed flow rate of 0.5 L min^{-1} and a transmembrane pressure of 1 bar .

Filtration Systems

BSA monomer-oligomer buffer optimization experiments were conducted using 50 mL stirred cell assemblies (Model 8050, Amicon Corporation, Beverly, MA) with flat membrane disks of 43 mm diameter. Stirred cells were pressurized with reagent-grade nitrogen, with the pressure monitored using a pressure transducer (Model TNF-R, Ohmeda Medical Devices Division, Inc., Madison, WI) and a digital monitor device (Model 610, Wedgewood Technology, Burlingame, CA). Stirred cell assemblies were placed on a multi-position stir plate (Model 725 C-P, Cole-Parmer Instrument Company, Niles, IL) and stirred at a constant rate of 450 rpm .

A Minitan-S[®] TFF holder (Millipore Corporation, Bedford, MA) with 60 cm^2 of membrane area and a fine screen (Cat. #PSSPFMT11) was used for the IgG-BSA buffer optimization and fluid dynamic optimization studies. The unit consists of a single membrane sheet placed between two gasketed screens in a Plexiglas holder. Because of the short channel length, the transmembrane pressure (ΔP) is approximately constant without the use of co-current filtrate flow. Feed flow rate was controlled with a gear pump (Pump Drive Motor Model 7144-00, $180\text{--}3600 \text{ rpm}$, Micro-pump Head Model 200-000, Cole-Parmer, Vernon Hills, IL) and measured with a flow meter (FR4000 series flow meters, Key Instruments, Trevose, PA) while filtrate flow rate was measured by timed collection of a known volume. Feed pressure was monitored using a pressure gauge ($0\text{--}2 \text{ bar}$ Test Gauge, U.S. Gauge, Sellersville, PA). The retentate pressure was controlled using a rising stem plug valve on the retentate line (Model 4F-PV4DE-V-SS, Parker, Jacksonville, AL).

BSA monomer-oligomer fluid dynamic optimization and two-stage closed-loop separation experiments were conducted in a fully automated scale down plant. The TFF systems consisted of two 40 L stainless steel tanks (feed/buffer and recycle) with a peristaltic transfer pump (Masterflex 7016, Cole-Parmer Instrument Company, Niles, IL) and a circumferential piston displacement feed pump (Model 6, Waukesha, Delavan, WI). Feed and filtrate in-line instrumentation included flow meters (Admag Model #102 and 105, Johnson Yokogawa Corporation, Newman, GA), pressure transducers (Model MSP220-A2, $0\text{--}100 \text{ psig} = 0\text{--}7 \text{ bar}$, Anderson Instruments, Fultonville, NY), and ultraviolet light detectors (Model 660, 280 nm , Wedgewood Technology, San Carlos, CA). The feed pipe also included a conductivity meter (Model 620, $0\text{--}100,000 \text{ }\mu\text{S}$, with a Model BT605-05-TL probe, Wedgewood Technology, San Carlos, CA) and a temperature probe (Model RIX, -29 to 82°C , Moore Industries, Sepulveda, CA). The retentate pipe included the same type of pressure transducers as used on the feed and filtrate lines. All piping was constructed of stainless steel, and all valves were pneumatically actuated

with Ethylene Propylene Diene Monomer diaphragms (Biotek Model #8836-18-BH, ITT Sherotec, Simi Valley, CA) except the retentate pressure control valve (Model #1/2 Mikroseal packless control valve, H.D. Baumann, Portsmouth, NY). Continuous tank liquid level was measured with a magneto restrictive probe (Model Tempsonics II, MTS, Research Triangle Park, NC). Data acquisition and control were performed using proprietary software (Genentech Inc., South San Francisco, CA) using a MycroAdvantage software shell (Moore Products, Springhouse, PA). Data acquisition was carried out at one second intervals for process control and monitoring, and at ten second intervals for historical data logging.

Stainless steel Pellicon-2™ holders, (Millipore Corporation, Bedford, MA) were used for both IgG-BSA and BSA monomer-oligomer separations. One to five cassettes can be assembled in a single mini Pellicon-2™ holder. Product enters through the bottom feed port and exists through the top retentate port. Filtrate was removed from both the top and bottom filtrate ports. The assembly nuts were manually torqued to 20 Nm corresponding to a force on the cassette sealing areas of 46 bar. Each mini-cassette has 0.1 m² of membrane area. Each cassette was manufactured with turbulence promoting screens on both the feed and filtrate sides of the membrane. The different feed screens (type A and C) employed in these cassettes differ with respect to their physical characteristics and allow for controlled channel dimensions. The type C screen has thicker filaments and a more open weave than the type A screen. All 10 kD regenerated cellulose membrane cassettes used the type C screen.

The HPTFF two-stage closed-loop cascade separation of BSA monomer from oligomers employed two of the previously described automated scale down filtration systems. A schematic diagram of the two-stage closed-loop cascade system is shown in Figure 1. The BSA monomer-oligomer feed pool was initially placed in the first stage tank. The first stage Pellicon-2™ holder contained a Biomax™ 150 kD membrane and was operated with co-current filtrate flow. The filtrate from the first stage was directed to the second

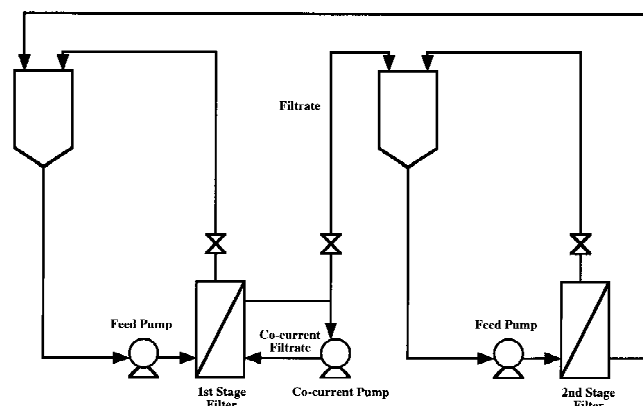


Figure 1. Schematic representation of a two-stage, closed-loop cascade system for protein separations using HPTFF.

stage feed tank. The second stage Pellicon-2™ holder contained a 10 kD regenerated cellulose membrane which was fully retentive to both BSA monomer and oligomers. The filtrate from the second stage, which was essentially pure buffer, was returned to the first stage tank. Tank liquid levels were maintained constant by controlling the filtrate flow rate in each stage at the same rate.

The IgG-BSA HPTFF separation was performed on a single stage manual Pellicon-2™ system with a Biomax™ 300 kD membrane. The experimental system consisted of three gear pumps, one each for feed, co-current filtrate, and diafiltration buffer addition (Pump Drive Motor Model 7144-00 180–3600 rpm, Micropump Head Model 200-000, Cole-Parmer, Vernon Hills, IL). Feed, filtrate, co-current filtrate, and diafiltration buffer flow rates were measured using in-line flow meters (FR4000 series flow meters, Key Instruments, Trevose, PA). Feed, retentate, and inlet and outlet filtrate pressures were measured using digital gauges (0–7 bar digital pressure gauge Model HG2000, Helicoid Instruments, Watertown, CT). Retentate pressure was controlled using a rising stem plug valve on the retentate line (Model 4F-PV4DE-V-SS, Parker, Jacksonville, AL). The feed pool containing the IgG-BSA mixture was placed in the system feed tank. A 20 min dynamic pre-conditioning of the membrane at gradually increasing filtrate flux was carried out in a total recirculation mode (retentate and filtrate cycled back to the feed tank). The IgG-BSA separation was performed in a constant retentate volume diafiltration mode. Feed samples were withdrawn periodically during the entire HPTFF separation for subsequent analysis.

Buffer Optimization Experiments

Buffer optimization for the BSA monomer-oligomer separation was performed using 50 mL Amicon stirred cells with 43 mm diameter Biomax™ 150 kD membranes. The experimental matrix consisted of six buffer compositions with ionic strengths of 10 and 50 mM and pH values of 4.0, 5.5, and 7.0. The buffering component of the pH 4.0 and pH 5.5 buffers was sodium acetate while the pH 7.0 solution was buffered with sodium phosphate. The freeze-dried BSA was dissolved in the appropriate buffer at a concentration of 5 g L⁻¹. A feed sample was taken for analysis and 50 mL of feed solution was placed in the stirred cell. Cells were pressurized with nitrogen and the first 2 mL of filtrate were used for flow rate determination. A subsequent 1 mL sample of the filtrate was collected and used for analysis. Filtrate flow rate was measured by timed collection of a known volume. Samples of feed and filtrate were analyzed by SEC.

Buffer optimization for the IgG-BSA separation was performed using a Minitan-S® system in a total recirculation mode (retentate and filtrate recycled to the feed tank). Eight sodium acetate buffers were examined with ionic strengths of 10 and 150 mM and pH values of 4.0, 4.8, 5.0, and 6.0. A matrix of 24 experiments was designed, consisting of the eight buffers and three membrane-pore sizes. The membranes investigated were all composite regenerated cellu-

lose membranes with nominal molecular weight cut-off values of 90, 130, and 170 kD. The feed solution (5 g L⁻¹ IgG and 0.2 g L⁻¹ BSA) was made by dissolving the proteins in the appropriate buffer solution. All experiments were conducted with a feed flow rate of 30 L h⁻¹ and a transmembrane pressure of 0.34 bar. The desired transmembrane pressure was set by adjusting the retentate pressure valve, and feed and filtrate samples were taken periodically for subsequent analysis. Feed and filtrate flow rates were also measured. Samples were assayed by capillary electrophoresis.

Fluid Dynamic and Membrane Optimization Experiments

The BSA monomer-oligomer fluid dynamic optimization study was conducted with 0.1 m² Pellicon-2™ cassettes on the automated scale down system. Biomax™ 150 kD membranes with either type A or C feed screens were used in experiments conducted in a total recirculation mode. BSA was dissolved in a 10 mM, pH 4.0 sodium acetate buffer at a concentration of 5 g L⁻¹. Experiments were conducted at three different feed fluxes corresponding to 192, 324, and 456 L m⁻² h⁻¹. For each feed flux, data were obtained at several transmembrane pressures, starting with low values and gradually increasing ΔP until the filtrate flux became relatively pressure independent. The system was allowed to stabilize for a minimum of 10 min at each ΔP before samples of the feed and filtrate were taken. The samples were analyzed by HPLC SEC.

The IgG-BSA membrane and fluid dynamic optimization study was performed with a Minitan-S® system using 60 cm² of flat sheet membrane. The membranes examined in the membrane optimization experiment were the regenerated cellulose 100 kD and the Biomax™ 300 kD. The membrane used for the fluid dynamic optimization study was the Biomax™ 300 kD. The experiments were conducted in total recirculation mode. The feed for these experiments consisted of a IgG-BSA mixture dissolved in a 10 mM, pH 4.0 sodium acetate buffer at a concentration of 5 g L⁻¹ IgG and 2 g L⁻¹ BSA. The feed flow rates examined in the fluid dynamic optimization were 0.30 and 0.75 L min⁻¹. Data and samples were collected at several transmembrane pressures as described above. Samples of the feed and filtrate were taken and assayed by HPLC anion exchange.

HPTFF Process Experiments

The BSA monomer-oligomer two-stage closed-loop cascade separations were performed on the automated scale down system. Three separate cascade experiments were conducted. All experiments were performed with a 10 mM, pH 4.0 sodium acetate buffer. The feed solution was 1.2–1.8 L of 5 g L⁻¹ crude BSA in the buffer. The first stage contained Pellicon-2™ cassettes with 0.2 m² of Biomax™ 150 kD membrane with type A screen, while the second stage contained Pellicon-2™ cassettes with 0.4–0.6 m² of 10 kD

regenerated cellulose membrane. The feed flux was 324 L m⁻² h⁻¹ for both stages and the first stage filtrate flux was controlled at set points of 130 or 165 L m⁻² h⁻¹ for the duration of the experiment. The second stage filtrate rate was controlled to be equal to the first stage filtrate rate.

The IgG-BSA HPTFF diafiltration separation was performed using the manual Pellicon-2™ system with 0.1 m² of Biomax™ 300 kD membrane with type A screen. The feed solution was 0.30 L of 5 g L⁻¹ IgG and 2 g L⁻¹ BSA monomer in a 10 mM, pH 4.0 sodium acetate buffer. The BSA monomer was obtained by HPTFF purification as described above. The system was run with a feed flux of 360 L m⁻² h⁻¹ and a filtrate flux of 120 L m⁻² h⁻¹. Diafiltration was carried out to an endpoint of 80 diavolumes.

Data Analysis

HPTFF process optimization and design is facilitated by use of the optimization diagram for membrane separations developed by van Reis and Saksena (1997). The optimization diagram uses two dimensionless parameters; Ψ (selectivity) and $N\Delta S$, both of which are used as parameters on a yield versus purification factor graph. The selectivity is defined as:

$$\Psi = S_1/S_2 \quad (1)$$

where S_1 and S_2 are the sieving coefficients for the two solutes ($S_1 > S_2$). The second dimensionless parameter is defined as:

$$N\Delta S = (JA t/V) (S_1 - S_2) \quad (2)$$

where N is the number of diavolumes, J is the filtrate flux, A is the membrane area, t is the diafiltration process time, and V is the retentate volume. It should be noted that A , t , and V are all elective parameters. Membrane area and time can be chosen based on engineering and economic constraints and the retentate volume can be altered by concentrating or diluting the feed. It is often useful to evaluate the experimental data before making a decision on these elective parameters. In order to specifically examine the effects of process variables on HPTFF performance, it is practical (and reasonable as a rule of thumb) to initially set:

$$At/V = 1 \quad (3)$$

where A has units of m², t is in hours, and V is in liters. Under these conditions $N\Delta S$ is proportional to $J\Delta S$, which is expressed only in terms of experimental parameters in units of L m⁻² h⁻¹:

$$J\Delta S = J (S_1 - S_2) \quad (4)$$

The yield and purification factor diagrams are constructed using the analysis presented by van Reis and Saksena (1997). Specifically, the yield and purification factor used to construct the optimization diagrams are given as:

$$Y_2 = e^{-NS_2} = e^{N\Delta S/(1-\Psi)} \quad (5)$$

$$P_2 = e^{N\Delta S} \quad (6)$$

where Y_2 and P_2 are the yield and purification factor for solute 2 in the retentate, and

$$Y_1 = 1 - e^{-NS_1} = 1 - e^{N\Delta S\Psi/(1-\Psi)} \quad (7)$$

$$P_1 = \frac{(1 - e^{-NS_1})/(1 - e^{-NS_2})}{[1 - e^{N\Delta S\Psi/(1-\Psi)}]/[1 - e^{N\Delta S/(1-\Psi)}]} \quad (8)$$

where Y_1 and P_1 are the yield and purification factor for solute 1 in the filtrate.

RESULTS

Buffer Optimization

Process optimization started with buffer selection. The goal of buffer optimization was to determine the best pH and ionic strength conditions for separation of the proteins of interest. A total of six buffer compositions were examined for the separation of BSA monomer from oligomers. The buffers were pH 4.0, 5.5, and 7.0 with ionic strengths of 10 and 50 mM. Experimental data for the monomer and oligomer sieving coefficients at a given solution pH and ionic strength were obtained over a range of filtrate fluxes. The optimum combinations of Ψ and $J\Delta S$ as a function of flux, for each buffer, were evaluated from the data with the results plotted as a function of pH in Figure 2A. There is a clear trade-off in this system between the optimum selectivity and $J\Delta S$. The best selectivity values are obtained at pH 7.0 while the maximum $J\Delta S$ values are obtained at pH 4.0. To determine the optimum buffer, it is necessary to consider the combination of these two factors and their effects on the yield and purification factor of the monomer species. The results of this calculation are shown for the pH 4.0 buffers in Figure 3A. The calculated (solid) lines are

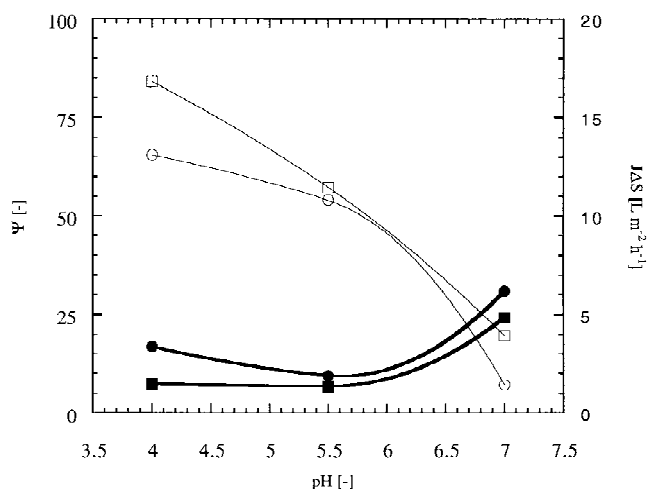


Figure 2A. Buffer optimization for BSA monomer-oligomer separation in stirred cells using Biomax™ 150 kD membrane; Ψ (filled markers) and $J\Delta S$ (open markers) as a function of buffer pH and ionic strength: 10 mM (●, ○) and 50 mM buffer (■, □). Buffers at pH values of 4.0 and 5.5 were sodium acetate while buffers at pH 7.0 were sodium phosphate. BSA was dissolved in the appropriate buffer at a concentration of 5 g L⁻¹.

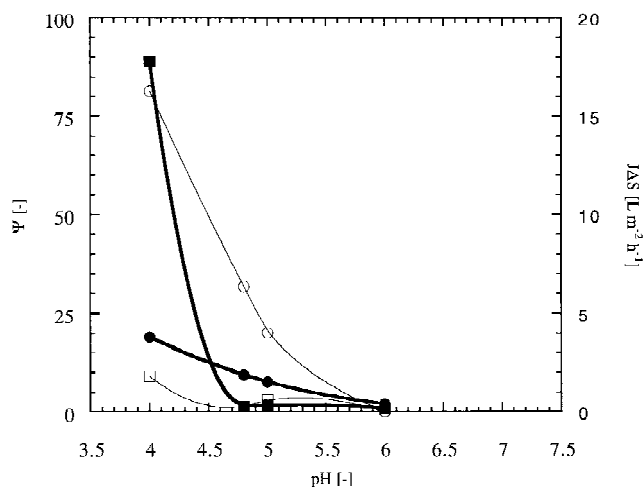


Figure 2B. Buffer optimization for IgG-BSA separation in Minitan-S® system with composite regenerated cellulose 170 kD membrane; Ψ (filled markers) and $J\Delta S$ (open markers) as a function of sodium acetate buffer pH and ionic strength: 10 mM (●, ○) and 150 mM buffer (■, □). IgG and BSA were dissolved in the appropriate buffer at 5 g L⁻¹ and 0.2 g L⁻¹ respectively.

superimposed on the optimization diagram for the yield and purification factor for a system where the product is in the filtrate. A limiting endpoint was determined by fixing the process time and surface area to volume ratio at $At/V = 1$ m² h L⁻¹. At a fixed value of At/V , the combination of the projected yield and purification factor was best for the 10 mM, pH 4.0 buffer. This buffer was chosen for the subsequent fluid dynamic optimization study for the BSA monomer-oligomer system. Alternatively, one could have chosen

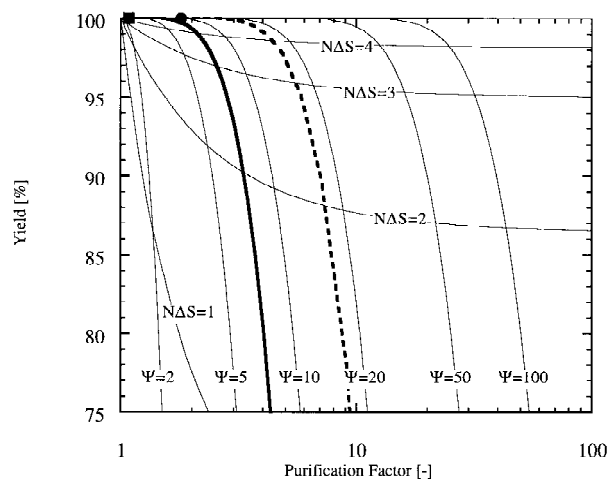


Figure 3A. Buffer optimization for BSA monomer-oligomer separation in the pH 4.0, sodium acetate buffers: 10 mM (---) and 50 mM buffer (—). Experimental conditions are the same as in Figure 2A. BSA monomer-oligomer selectivity values and maximum $N\Delta S$ (for $At/V = 1$ m² h L⁻¹) obtained from the experimental data for Ψ and $J\Delta S$ in Figure 2A. A process with product in the filtrate can be run along the appropriate selectivity line from 0% yield anywhere up to the $N\Delta S$ endpoint (shown by the markers). The thin lines represent the theoretical yield and purification factors calculated from Equations (7) and (8) for an optimization diagram with the product in the filtrate.

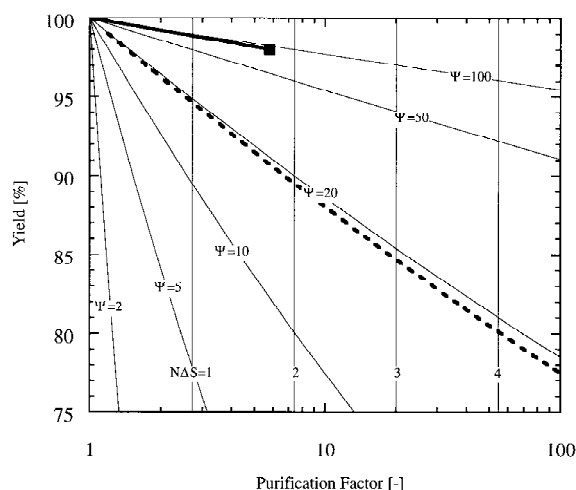


Figure 3B. Buffer optimization for IgG-BSA separation in the pH 4.0 sodium acetate buffers: 10 mM (---○---); and 150 mM buffer (—●—). Experimental conditions are the same as in Figure 2B. IgG selectivity values and maximum $N\Delta S$ (for $At/V = 1 \text{ m}^2 \text{ h L}^{-1}$) obtained from the experimental data for Ψ and $J\Delta S$ in Figure 2B. A process with product in the retentate can be run along the appropriate selectivity line from 100% yield anywhere up to the $N\Delta S$ endpoint (shown by the markers). The theoretical limiting endpoint for the 10 mM buffer (off scale) was at <75% yield and a purification factor >100. The thin lines represent the theoretical yield and purification factors calculated from Equations (5) and (6) for an optimization diagram with the product in the retentate.

the 10 mM, pH 7.0 buffer, which has a higher selectivity but a lower $J\Delta S$. This would enable a higher purification factor but a lower yield for the same At/V ratio. By increasing At/V , however, one could accomplish both a higher purification factor and yield if this could be justified for this particular separation within economic and engineering constraints.

A total of eight buffers were investigated for the IgG-BSA separation. The buffers were pH 4.0, 4.8, 5.0, and 6.0 with ionic strengths of 10 and 50 mM. A matrix of 24 experiments consisting of the eight buffers and three composite regenerated cellulose membranes (90, 130, and 170 kD) was designed. Each experiment was carried out in duplicate, with the data reported in Figure 2B representing an average of both runs. To illustrate the buffer effects, results with only one type of membrane (170 kD composite regenerated cellulose) are presented in this figure. Similar results were obtained with the other membranes. Figure 2B shows that both the selectivity and $J\Delta S$ increase with decreasing pH. This behavior is due to the reduction in BSA retention at lower pH associated with the smaller net charge on BSA near its isoelectric point (pH 4.8). This effect has been discussed previously by Saksena and Zydney (1994). Experimental values for the sieving coefficients and filtrate flux were used to project the yield and purification factors of IgG as a function of the number of diavolumes. The results of this calculation are shown for the pH 4.0 buffers in Figure 3B. These lines are superimposed on the general optimization diagram for a system where the product is in the retentate. For a fixed value of $At/V = 1 \text{ m}^2 \text{ h L}^{-1}$, the 150 mM,

pH 4.0 buffer predicted the highest yield of IgG at the expense of a lower purification factor. Conversely, the use of a 10 mM, pH 4.0 buffer resulted in a higher projected purification factor with a lower yield of IgG. This buffer was chosen for the subsequent fluid dynamic optimization study to test the purification capability of HPTFF.

Fluid Dynamic and Membrane Optimization

The next step in process optimization was selection of optimum fluid dynamic conditions. The fluid dynamic optimization study examined the effects of feed flux, transmembrane pressure, and the resulting filtrate flux on the separation of BSA monomer from oligomers. Effects of turbulence promotion in the cassettes were also examined with type A and type C screens. Pellicon-2™ holders with Biomax™ 150 kD cassettes were used in this study. The feed solution was crude BSA at a concentration of 5 g L^{-1} in 10 mM, pH 4.0 sodium acetate, based on the results of the buffer optimization study. These experiments were conducted in total recirculation mode using the automated scale down system. At each of the feed fluxes of 192, 324, and $456 \text{ L m}^{-2} \text{ h}^{-1}$ a ΔP excursion was performed as described earlier. The system was allowed to stabilize at each pressure set point before samples of the feed and filtrate were taken for subsequent HPLC SEC analysis.

The experimental data for the sieving coefficients of the BSA monomer and oligomer were used to calculate the selectivity and $J\Delta S$ values with the results plotted in Figures 4A and 4B. Figure 4A represents the results of the separation using the type A screen while the corresponding results for the type C screen are shown in Figure 4B. These figures show that there is an optimum selectivity and $J\Delta S$ for each feed flux rate. However, the optimum selectivity does not

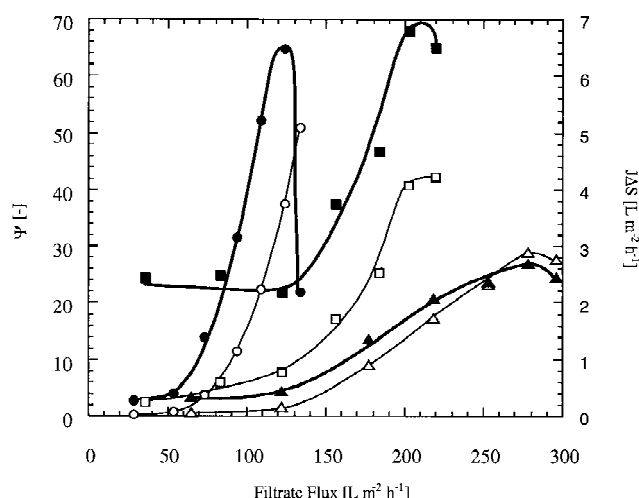


Figure 4A. Fluid dynamic optimization for BSA monomer-oligomer separation with type A screen Pellicon-2™ cassette, with Biomax™ 150 kD membrane. BSA concentration of 5 g L^{-1} in 10 mM, pH 4.0 sodium acetate buffer; Ψ (filled markers) and $J\Delta S$ (open markers) as a function of filtrate flux at three different feed flux rates: $192 \text{ L m}^{-2} \text{ h}^{-1}$ (●, ○); $324 \text{ L m}^{-2} \text{ h}^{-1}$ (■, □); and $456 \text{ L m}^{-2} \text{ h}^{-1}$ (▲, △).

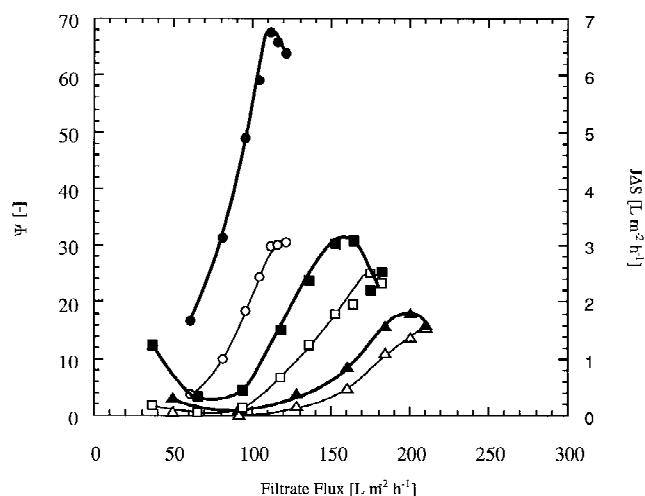


Figure 4B. Fluid dynamic optimization for BSA monomer-oligomer separation with type C screen Pellicon-2™ cassette, with Biomax™ 150 kD membrane. BSA concentration of 5 g L⁻¹ in 10 mM, pH 4.0 sodium acetate buffer; Ψ (filled markers) and JΔS (open markers) as a function of filtrate flux at three different feed flux rates: 192 L m⁻² h⁻¹ (●, ○); 324 L m⁻² h⁻¹ (■, □); and 456 L m⁻² h⁻¹ (▲, △).

always occur at the same fluid dynamic conditions as the optimum JΔS. In this case, the data indicate that the results with type A and type C screens were similar within the filtrate flux range studied with both screens. The experiments with type A screen, however, were extended to higher filtrate fluxes and resulted in higher optimum selectivity and JΔS combinations. The optimum selectivity and JΔS combinations at each feed flux rate were used to project yield and purification factor as described in the buffer optimization results. The results of this calculation for the type A screen cassette are shown in Figure 5A, with the feed flux rate of either 192 or 324 L m⁻² h⁻¹ producing the best yield and purification factor combination.

In the separation of IgG and BSA, the fluid dynamic optimization was conducted in a Minitan-S® system. The effects of feed flux, transmembrane pressure, and resulting filtrate flux on the separation of IgG and BSA were examined using the Biomax™ 300 kD membrane. The effect of membrane chemistry was also examined with experiments performed using both regenerated cellulose 100 kD and Biomax™ 300 kD membranes. The 10 mM, pH 4.0 sodium acetate buffer was selected from the buffer optimization study. The protein solution was a IgG-BSA mixture with a concentration of 5 g L⁻¹ IgG and 2 g L⁻¹ BSA. The fluid dynamic optimization experiments were conducted in total recirculation mode at feed rates of 0.30 and 0.75 L min⁻¹. The membrane optimization experiments were conducted in total recirculation mode at a feed rate of 0.3 L min⁻¹. In both cases, the system was allowed to equilibrate for 20 min at each ΔP set point before samples of the feed and filtrate were taken for subsequent analysis. The experimental data for the sieving coefficients were used to calculate the selectivity and JΔS values at each filtrate flux. The optimum selectivity at each feed flux rate, and the corresponding

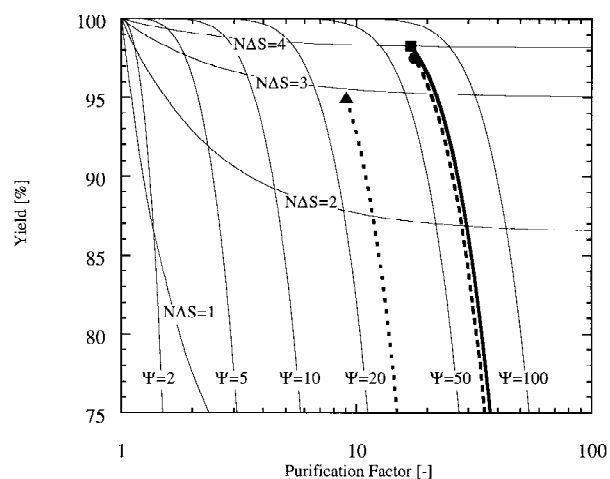


Figure 5A. Fluid dynamic optimization for BSA monomer-oligomer separation with type A screen Pellicon-2™ cassette, with Biomax™ 150 kD membrane. BSA concentration of 5 g L⁻¹ in 10 mM pH = 4.0 sodium acetate buffer. BSA monomer-oligomer selectivity values and maximum NΔS (for At/V = 1 m² h L⁻¹), were obtained from the experimental data for the best combination of Ψ and JΔS in Figure 4A, at each of the feed fluxes of: 192 L m⁻² h⁻¹ (●); 324 L m⁻² h⁻¹ (■); and 456 L m⁻² h⁻¹ (▲).

measured filtrate flux, were used to project yield and purification factor as previously described. The results of the fluid dynamic optimization are shown in Figure 5B. The results at both feed flow rates were very similar. The results of the membrane optimization experiment are shown in Figure 5C. The Biomax™ 300 kD membrane had higher se-

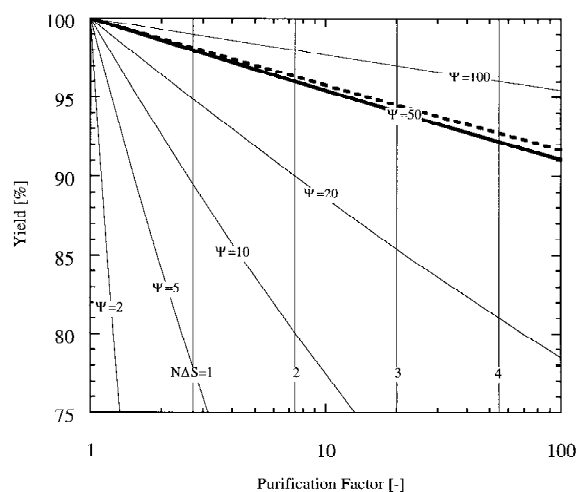


Figure 5B. Fluid dynamic optimization for IgG-BSA separation in Minitan-S® system, with Biomax™ 300 kD membrane. IgG and BSA concentration of 5 g L⁻¹ and 2 g L⁻¹ respectively, in 10 mM, pH = 4.0 sodium acetate buffer. IgG selectivity values and maximum NΔS (for At/V = 1 m² h L⁻¹), were obtained from the experimental data for the best combination of Ψ and JΔS obtained in the fluid dynamic optimization experiment, at each of the feed flow rates of: 0.30 L min⁻¹ (—); and 0.75 L min⁻¹ (---). The theoretical limiting endpoint for the 0.3 L min⁻¹ feed rate case (off scale) was 85% yield and a purification factor of 6600, and the 0.75 L min⁻¹ feed rate case limiting endpoint (off scale) was 84% yield and a purification factor of 5300.

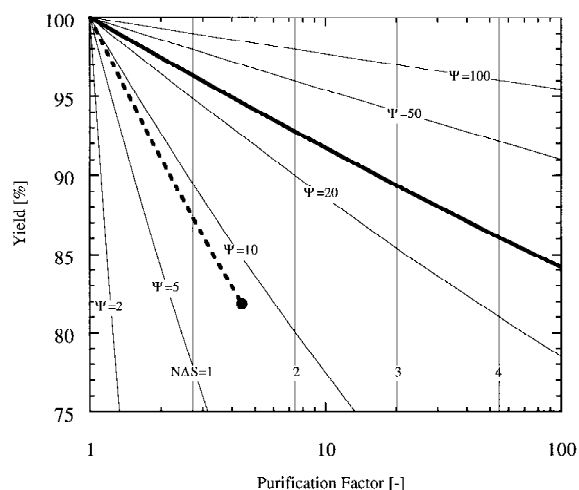


Figure 5C. Membrane optimization for IgG-BSA separation in Minitan-S® system at a feed flow rate of 0.30 L min^{-1} . IgG and BSA concentration of 5 g L^{-1} and 2 g L^{-1} respectively, in 10 mM sodium acetate buffer. IgG selectivity values and maximum $N\Delta S$ (for $At/V = 1 \text{ m}^2 \text{ h L}^{-1}$), were obtained from the experimental data for the best combination of Ψ and $J\Delta S$ obtained in a fluid dynamic optimization experiment, for each membrane chemistry: conventional Regenerated Cellulose 100 kD (—); and Biomax™ 100 kD (—). The theoretical limiting endpoint for the Biomax™ 100 kD case (off scale) was at $<75\%$ yield and a purification factor >100 .

lectivity than the regenerated cellulose, which results in a much better predicted yield and purification factor.

HPTFF Processes

The information gained from the buffer and fluid dynamic optimization studies were used to select optimum operating conditions for the HPTFF two-stage closed-loop cascade separation of BSA monomer from oligomers. The two-stage system allows for simultaneous purification and concentration of the BSA monomer. The closed-loop configuration provides an opportunity to perform extended HPTFF diafiltration without the need for large volume buffer consumption. An extensive number of diavolumes can hence be achieved with a buffer volume that equals the protein feed volume. The large number of diavolumes is therefore not a concern for practical implementation of these processes. The two-stage closed-loop cascade separation of BSA monomer from oligomers was performed on the automated scale down system. The initial feed solution was crude BSA at a concentration of 5 g L^{-1} in $\text{pH } 4.0$ sodium acetate buffer at 10 mM ionic strength.

Experiments were attempted at filtrate fluxes of 130 , 165 and $200 \text{ L m}^{-2} \text{ h}^{-1}$. A constant flux of $200 \text{ L m}^{-2} \text{ h}^{-1}$ could not be maintained throughout the process without significant increases in transmembrane pressure. This was likely due to long-term fouling effects which would not be detected in the fluid dynamic optimization experiments. The chromatograms of the initial crude BSA pool and the final second stage (filtrate) pool for the $165 \text{ L m}^{-2} \text{ h}^{-1}$ process are shown in Figure 6A. In this case, the separation resulted in

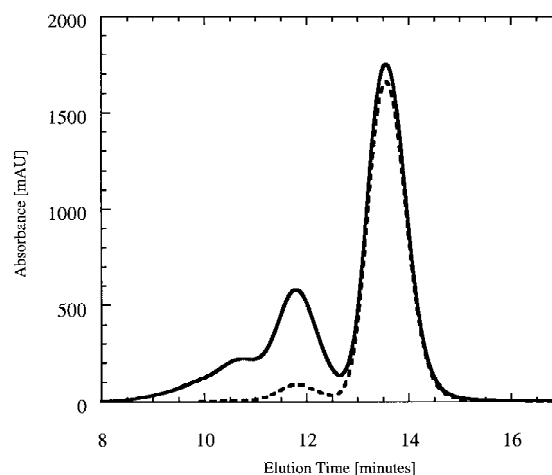


Figure 6A. Two-stage closed-loop cascade HPTFF separation of 5 g L^{-1} BSA monomer-oligomer, in 10 mM $\text{pH } 4.0$ sodium acetate buffer, performed with type A screen Pellicon-2™ cassette with Biomax™ 150 kD membrane. A yield of 86% and a purification factor of 9 were achieved with an $At/V = 1.07 \text{ m}^2 \text{ h L}^{-1}$ using 179 diavolumes in a process time of 9 h . HPLC SEC chromatograms of: initial feed pool (—); final second stage (filtrate) pool (-----).

a filtrate pool (second stage) of purified BSA monomer and a retentate pool (first stage) of BSA containing oligomers and some residual monomer. The yield of BSA monomer in the final pool was 86% and the purification factor was 9 . This was achieved with 179 diavolumes in 9 h with a At/V ratio of $1.07 \text{ m}^2 \text{ h L}^{-1}$ and a filtrate flux of $165 \text{ L m}^{-2} \text{ h}^{-1}$.

A comparison of the predicted and actual process yields and purification factors is shown in Table I. Predictions were performed using Equations (7) and (8) with the selectivity and $J\Delta S$ determined from the fluid dynamic optimization data in Figure 4A for the purification of BSA mono-

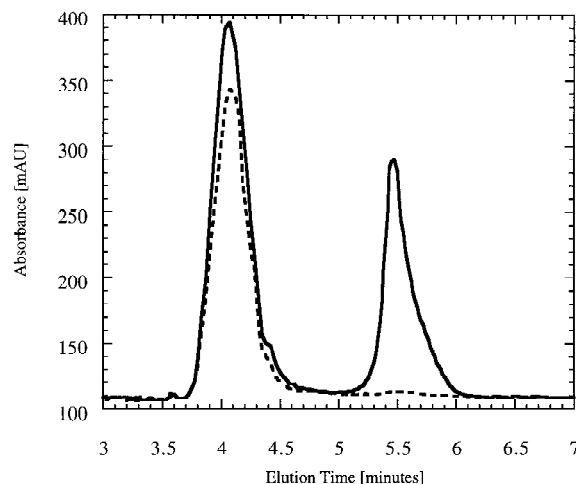


Figure 6B. Single stage HPTFF separation of IgG-BSA performed in the Pellicon-2™ system with Biomax™ 300 kD membrane, 5 g L^{-1} IgG and 2 g L^{-1} BSA in 10 mM $\text{pH } 4.0$ sodium acetate buffer. A yield of 84% and a purification factor of 30 were achieved with an $At/V = 0.67 \text{ m}^2 \text{ h L}^{-1}$ using 80 diavolumes in a process time of 3.3 h . Capillary electrophoresis analysis of: initial feed pool (—); final retentate pool (-----).

Table I. Comparison of predicted and actual process purification factors.

	Diavolumes $N (-)$	Yield $Y (%)$	Purification factor $(-)$
Process ($130 \text{ L m}^{-2} \text{ h}^{-1}$)	80	51	32
Prediction (Equal N)	80	47	19
Prediction (Equal Y)	89	51	18
Process ($165 \text{ L m}^{-2} \text{ h}^{-1}$)	77	69	10
Prediction (Equal N)	77	59	26
Prediction (Equal Y)	101	69	23
Process ($165 \text{ L m}^{-2} \text{ h}^{-1}$)	179	86	9
Prediction (Equal N)	179	87	17
Prediction (Equal Y)	168	86	18

Predictions are based on the fluid dynamic optimization data in Figure 4A for the purification of BSA monomer from a BSA feed stock in a 10 mM pH 4.0 acetate buffer. The fluid dynamic conditions were a feed flux of $324 \text{ L m}^{-2} \text{ h}^{-1}$ and filtrate fluxes of $130 \text{ L m}^{-2} \text{ h}^{-1}$ and $165 \text{ L m}^{-2} \text{ h}^{-1}$. Predictions were made for both equal number of diavolumes and equal yield compared to the actual cascade HPTFF process (Fig. 6A).

mer from a BSA feed stock in a 10 mM pH 4.0 acetate buffer at a feed flux of $324 \text{ L m}^{-2} \text{ h}^{-1}$ and filtrate fluxes of 130 and $165 \text{ L m}^{-2} \text{ h}^{-1}$. Calculations were done with either the yield or the number of diavolumes chosen so as to match the data from the two-stage cascade separation. The data shows good agreement in yields. The purification factors in the HPTFF process were either higher or lower than those predicted by the fluid dynamic optimization experiments. This indicates that $J\Delta S$ values achieved during the HPTFF process were consistent with the fluid dynamic optimization data whereas the selectivity was either lower or higher than expected. Further work is required to study and control process variability.

An HPTFF separation of IgG and BSA monomer was also performed to yield a pool of purified IgG. The protein solution was 5 g L^{-1} IgG and 2 g L^{-1} BSA monomer in a 10 mM, pH 4.0 sodium acetate buffer. Results from the capillary electrophoresis analysis of the initial mixed IgG-BSA pool, and the final purified IgG pool, are shown in Figure 6B. The yield of IgG in the final pool was 84% and the purification factor was 30. This was achieved with 80 diavolumes in 3.3 h with a At/V ratio of $0.67 \text{ m}^2 \text{ h L}^{-1}$ and a filtrate flux of $120 \text{ L m}^{-2} \text{ h}^{-1}$.

DISCUSSION

The results obtained in this study clearly demonstrate that HPTFF can be used for protein separations even when the species involved have similar molecular weights. High Ψ and $J\Delta S$ values were obtained for both BSA monomer-oligomer and IgG-BSA separations by optimizing the membrane pore size, membrane chemistry, the solution (buffer) chemistry, and the fluid dynamics. The best combination of Ψ and $J\Delta S$ was determined using appropriate optimization diagrams (van Reis and Saksena, 1997) for these membrane separations. The actual protein purification procedures were

performed in a diafiltration mode to effectively wash out the more permeable species. In one of the applications the total buffer consumption was drastically reduced through the use of a two-stage closed-loop cascade system. The net result was the development of HPTFF systems which had high yields and purification factors for BSA monomer-oligomer and IgG-BSA separations.

The high Ψ and $J\Delta S$ values that were obtained with the BSA monomer-oligomer system were actually quite unexpected. These species have essentially identical surface charge densities, and the dimer and tetramer differ from the monomer in hydrodynamic radius by less than 30% and 60%, respectively, assuming a 1/3 power law relationship with the molecular weight. Previous studies of selective membrane filtration indicate that it should not be possible to obtain any significant separation between these species. Yet the actual selectivity values obtained in this study were approximately 20, resulting in a purification factor of 9 and a monomer yield of 86% in the two-stage closed-loop cascade system.

The high selectivity for the BSA monomer-oligomer system was a direct result of the very high retention of the less permeable species, with the oligomer retention being greater than 99.98% at the optimal conditions. The significant passage of monomer under these conditions was surprising, but this behavior is likely due to the combined electrostatic and steric interactions between the proteins and the membrane pores. The electrostatic interactions between the protein and the membrane depend strongly on their surface charge densities. The surface charge densities of pure BSA monomer and dimer were estimated from their respective electrophoretic mobility values, with the latter measured experimentally using standard capillary electrophoresis procedures. The surface charge densities of both the monomer and the dimer were found to be essentially identical, with the surface charge density of the dimer being slightly larger than that of the monomer. However, the electrostatic interactions between the oligomers and the pore boundaries can be much greater than that for the monomer due to the much smaller distance between the surface of the oligomers and the pore. The net result is an increase in oligomer retention relative to that of the monomer. Additional experimental studies will be required to fully elucidate the separation mechanism in this type of monomer-oligomer system.

Although the detailed optimization of HPTFF is likely to be somewhat different for every protein system, the results obtained in this study, in combination with results from previous studies of selective protein filtration, provide important insights into how to select the optimal conditions for effective protein separations. To optimize yields and purification factors in HPTFF it is necessary to optimize both Ψ and $J\Delta S$ (van Reis and Saksena, 1997). The high selectivity can be most easily achieved by using a membrane that has a very high retention of one of the proteins (i.e., $R_2 \approx 1$), which in turn requires a membrane that is essentially free of any very large pores or defects. This was achieved in this

study using newly developed modified polyethersulfone membranes. Future work will explore in more detail the criteria for membrane selection in HPTFF.

The high protein retention can be significantly enhanced by exploiting electrostatic interactions between the protein and membrane pores as discussed by Saksena and Zydney (1994). This was in fact the basis for the high selectivity achieved with the IgG-BSA system examined in the current study. IgG rejection at pH 4.0 was very high in the low ionic strength buffer due to the strong electrostatic exclusion of the positively charged IgG from the membrane pores. The high $J\Delta S$ value needed for effective protein separations requires both a moderately high filtrate flux and a reasonable passage of the more permeable species.

To obtain the necessary protein passage, it is desirable to operate the system at a pH near the isoelectric point of the more permeable protein to minimize the electrostatic exclusion of this solute. This strategy was exploited by Saksena and Zydney (1994) in their study of IgG-BSA separations and by van Eijndhoven et al. (1995) in their analysis of BSA-Hemoglobin purification.

It is also advantageous to operate under conditions that encourage high degrees of concentration polarization to increase the overall protein passage even when the intrinsic protein transmission is quite small. This approach was first suggested by Nakatsuka and Michaels (1992) in their analysis of BSA-Myoglobin separation by dead-end (unstirred) filtration. To properly exploit this phenomenon in a TFF system, it is necessary to have a uniform degree of concentration polarization throughout the tangential flow module. This requires careful control of the filtrate flux and the local transmembrane pressure, which can be achieved by operating the system with a co-current filtrate flow to effectively balance the axial pressure drop on the retentate side (van Reis, US Patent #5,256,294, 1993). Note that it is even possible to use the concentration polarization phenomenon to obtain reverse selectivity values in HPTFF, with a higher passage of the larger molecular weight (smaller mass transfer coefficient) species. This effect has been demonstrated experimentally using both polyethylene glycols (Jonsson, 1986) and IgG-BSA mixtures (Saksena and Zydney, 1994).

HPTFF has the potential to provide an important new unit operation for the downstream processing of macromolecular species. The data obtained in this study demonstrate that HPTFF provides purification factors and yields for BSA monomer-oligomer and IgG-BSA separations which are comparable to yields and purification factors for a range of commercial protein separation processes. HPTFF also has the potential to provide higher throughput than SEC, analogous to previous comparisons of TFF and SEC for buffer exchange of proteins (Kurnik et al., 1995). Further studies will be required to directly compare yield, purification, throughput, and the economics of these two unit operations. It should be noted that HPTFF separations can exploit both size and charge differences between proteins.

In this study, we demonstrated the successful separation of BSA monomer from oligomers with the same isoelectric

point. Van Eijndhoven et al. (1995) previously demonstrated the separation of hemoglobin (MW = 67 kD, pI = 6.85) from BSA (MW = 69 kD, pI = 4.7). In this study, we also demonstrated a very high resolution separation of BSA from IgG in which both size and charge differences were exploited. It will, therefore, be of great interest to compare HPTFF performance with ion exchange chromatography. HPTFF systems are also easily scaled, with recent ultrafiltration studies demonstrating linear scale-up over a 1,000-fold scale using properly designed membrane systems. HPTFF systems are also unique in that multiple steps can, at least in principle, be performed using a single unit operation. This includes buffer exchange, protein concentration, and protein separation, all of which can be performed in an HPTFF system using appropriate membranes and buffer combinations. Multiple purifications can be envisioned by using sequential steps in which buffer composition and fluid dynamics are changed to affect the selective removal of various impurities. Although the ultimate range of HPTFF applications in bioprocessing (and related fields) remains to be determined, this membrane technology clearly has the potential to play a vital role in the development of new and improved separation processes for many macromolecular species.

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NOMENCLATURE

A	membrane area (m^2)
BSA	bovine serum albumin
HPLC	high pressure liquid chromatography
HPTFF	high performance tangential flow filtration
IgG	immunoglobulin G
J	filtrate flux ($\text{L m}^{-2} \text{h}^{-1}$)
MW	molecular weight (kD)
N	number of diavolumes (-)
P_i	purification factor for solute i (-)
ΔP	transmembrane pressure (bar)
S_i	sieving coefficient for solute i (-)
ΔS	difference in solute sieving coefficients (-)
SEC	size exclusion chromatography
t	process time (h)
TFF	tangential flow filtration
V	retentate volume during constant volume diafiltration (L)
Y_i	yield of solute i (-)

Greek Letters

Ψ	Selectivity (-)
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Subscripts

1	Solute 1
2	Solute 2

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